



CLOUD COMPUTING

Complete Study Notes



Topics Covered in This Document

Chapter 1 — What is Cloud?	Chapter 2 — Public vs Private Cloud
Chapter 3 — Virtualization	Chapter 4 — AWS vs Others
Chapter 5 — IAM in AWS	Chapter 6 — Practical: S3 + IAM Lab



CHAPTER 1 — What is Cloud?



Simple Explanation (Student Language)

Imagine you have 100 photos on your phone. Your phone storage gets full. So you upload them to Google Photos — now your photos are stored somewhere on the internet, and you can access them from ANY device, ANYWHERE.



That 'somewhere on the internet' = THE CLOUD

You don't need to carry your files. You don't need to worry about storage. They are always there — safe, accessible, from anywhere!



Technical Definition



Definition:

The Cloud is a network of remote servers hosted on the internet that store, manage, and process data — instead of using a local computer or hard drive.



Real Life Analogy — The Electricity Example

Think of Cloud like ELECTRICITY from a power grid.

- ⚡ You don't build your own power plant at home.
- ⚡ You just PLUG IN and use electricity.
- ⚡ You PAY ONLY for what you use.

Similarly — companies don't build their own servers.

They just plug into the Cloud (AWS, Azure, Google Cloud) and pay for what they use!

Real Life Examples of Cloud

You Use	What Happens in the Cloud
Google Drive	Your files stored on Google's servers worldwide
Netflix	Movies stored and streamed from Amazon AWS servers
WhatsApp	Messages backed up on Meta's cloud servers

3 Types of Cloud Services (Technical)

Type	Full Name	What It Means + Example
IaaS	Infrastructure as a Service	You RENT servers, storage, networking. Example: Amazon AWS EC2, Rackspace
PaaS	Platform as a Service	You rent a PLATFORM to build apps. Example: Google App Engine, Heroku
SaaS	Software as a Service	You use READY-MADE software online. Example: Gmail, Zoom, Dropbox, Canva

Why Cloud is Powerful

Benefit	Explanation
 Access Anywhere	Use any phone, laptop, or tablet from any location
 Cost Saving	No need to buy expensive hardware or servers
 Scalable	Need more storage? Upgrade in seconds — no hardware buy
 Backup & Security	Data is safe even if your phone breaks or gets stolen
 Speed	Powerful remote servers process faster than your laptop



CHAPTER 2 — Public Cloud vs Private Cloud

Simple Explanation (Bus vs Car Analogy)

Public Cloud = A BUS

- Many people share it
- Cheap & available for everyone
- You don't own it

Private Cloud = YOUR CAR

- Only YOU use it
- More expensive, fully yours
- Complete control & privacy

What is Public Cloud?

Definition:

Public Cloud is a cloud environment where the infrastructure (servers, storage, networking) is OWNED and MANAGED by a third-party provider and SHARED among MULTIPLE customers over the internet.

Public Cloud Real Life Examples

Service You Use	Cloud Provider Behind It
Gmail / Google Drive	Google Cloud Platform
Netflix	Amazon AWS
LinkedIn	Microsoft Azure
Dropbox	Amazon AWS
Zoom	Oracle Cloud + AWS

What is Private Cloud?

Definition:

Private Cloud is a cloud environment EXCLUSIVELY used by ONE organization.

The infrastructure is either hosted on-premise (inside their own building) or by a third party — but it is NEVER SHARED with anyone else.

Private Cloud Real Life Examples

Organization	Why They Use Private Cloud
Banks (SBI, HDFC)	Customer financial data is highly sensitive
Indian Army / DRDO	Defence data cannot be on public servers
Hospitals (AIIMS)	Patient medical records need strict privacy
Indian Government	Citizen data must stay within the country

Public Cloud vs Private Cloud — Full Comparison

Feature	 Public Cloud	 Private Cloud
Ownership	Third Party (AWS, Google)	Your Company owns it
Users	Many companies share it	Only YOU use it
Cost	Low — pay per use	High — big investment
Security	Good	Very High
Control	Limited	Full Control
Setup Time	Minutes	Months
Scalability	Instantly scalable	Takes time and money
Best For	Startups, small business	Banks, Govt, Defence

BONUS — Hybrid Cloud (Best of Both!)

Many big companies use BOTH — this is called HYBRID CLOUD

 Sensitive data → Private Cloud

 Normal operations → Public Cloud

Example: A bank keeps customer account data on Private Cloud, but runs its website and mobile app on Public Cloud (AWS/Azure).



CHAPTER 3 — Virtualization



Simple Explanation

Imagine you have ONE powerful laptop but you need to run Windows for gaming, Linux for coding, and macOS for designing.

Do you buy 3 separate laptops? That's expensive! Instead — you use Virtualization to run ALL 3 operating systems on the SAME laptop at the same time!



Technical Definition



Definition:

Virtualization is the process of creating a software-based (virtual) version of something — like a server, storage, network, or operating system — using a special software called a HYPERVISOR, that divides ONE physical machine into multiple Virtual Machines (VMs).



Real Life Analogy — The Apartment Building



Think of a HUGE EMPTY BUILDING (Physical Server)

A builder (Hypervisor) comes and divides it into MULTIPLE FLATS (Virtual Machines):

Flat 1 → Family A lives here 🏠

Flat 2 → Family B lives here 🏠

Flat 3 → Family C lives here 🏠

Each family has their OWN space, privacy, and independence — but they all share the SAME building, electricity, and water.

Nobody interferes with each other — just like VMs!

5 Types of Virtualization

Type	What It Does	Real Example
1. Server Virtualization	One physical server → Multiple virtual servers	Amazon AWS EC2 instances
2. Desktop Virtualization	Run multiple OS on one laptop/desktop	VirtualBox, VMware on your PC
3. Network Virtualization	Create virtual networks inside physical ones	VPN (Virtual Private Network)
4. Storage Virtualization	Combine multiple physical storages into one	Google Drive (1 drive feel, 1000s of disks)
5. App Virtualization	Run an app without installing it	Google Docs (runs in browser)

Types of Hypervisor

TYPE 1 — Bare Metal Hypervisor Runs DIRECTLY on hardware — No OS needed. Very FAST and used in big data centers. Flow: Hardware → Hypervisor → VMs Examples: VMware ESXi, Microsoft Hyper-V	TYPE 2 — Hosted Hypervisor Runs ON TOP of an existing OS. Used on personal computers and laptops. Flow: Hardware → Host OS → Hypervisor → VMs Examples: VirtualBox, VMware Workstation
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Benefits of Virtualization

Benefit	Explanation
 Cost Saving	Buy 1 server instead of 10. Less electricity, less space needed
 Better Resource Usage	A server sitting idle is waste. Virtualization uses 100% of its power
 Isolation & Security	If VM1 crashes or gets a virus, VM2 and VM3 are completely safe
 Easy Backup & Recovery	Take a Snapshot of VM. If something breaks, restore in seconds
 Scalability	Need more VMs? Create them in minutes — no new hardware required

Virtualization & Cloud — The Connection

☛ CLOUD IS BUILT ON TOP OF VIRTUALIZATION!

When you rent a server on AWS or Azure — you are actually getting a VIRTUAL MACHINE running on their massive physical servers.

Your AWS EC2 Instance (VM)



Amazon's Hypervisor



Amazon's Physical Server (in a data center in Mumbai/USA)

🔗 **Virtualization = Dividing ONE powerful physical machine into MULTIPLE virtual machines using a Hypervisor — saving cost, increasing efficiency, and giving full isolation.**



CHAPTER 4 — How AWS is Better Than Others



Simple Explanation — Restaurant Analogy

Imagine you want to open a restaurant and need kitchen equipment suppliers:

- 🏆 1 AWS (Amazon) = The biggest, oldest, most trusted supplier — has EVERYTHING
- 🥈 2 Azure (Microsoft) = Great if you already use Microsoft products
- 🥉 3 Google Cloud = Amazing for AI & Data — but smaller network

AWS came FIRST, grew the MOST, and still leads the market!



Market Share — Who Rules the Cloud?

Cloud Provider	Market Share	Position
🏆 AWS (Amazon)	31%	#1 for 13+ consecutive years
🥈 Azure (Microsoft)	25%	#2 — strong in enterprise
🥉 Google Cloud (GCP)	11%	#3 — best in AI/Data
Others Combined	33%	Alibaba, IBM, Oracle, etc.



History — Why AWS Has an Advantage

2006 → AWS Launched ★ (FIRST mover advantage)

2010 → Azure Launched (4 years LATE)

2011 → Google Cloud Launched (5 years LATE)

AWS had 4-5 years HEAD START to:

- ☑ Build more data centers worldwide
- ☑ Gain millions of customers
- ☑ Develop 200+ services
- ☑ Earn the trust of big organizations

Advantage 1 — Most Services

Category	AWS 🛠️ vs Competitors
Compute	EC2 🛠️ vs Azure VMs vs Compute Engine
Storage	S3 🛠️ vs Azure Blob Storage vs Cloud Storage
Database	RDS + DynamoDB 🛠️ vs Azure SQL vs Cloud SQL
AI / ML	SageMaker vs Azure ML vs Vertex AI 🛠️ (GCP wins)
Serverless	Lambda 🛠️ vs Azure Functions vs Cloud Functions
Total Services	200+ 🛠️ vs 170+ vs 150+

Advantage 2 — Security & Trust

WHO TRUSTS AWS FOR THEIR MOST SENSITIVE DATA?

-  NASA (Space agency)
-  Goldman Sachs, JP Morgan (Global banks)
-  NHS UK (National Health Service)
-  CIA, Pentagon (US Government Defence)
-  Netflix, Disney+ (Media giants)
-  Flipkart, Zomato (Indian companies)

AWS has 143+ security compliance programs —
MORE than Azure and Google Cloud combined!

Advantage 3 — Pricing Flexibility

Pricing Model	What It Means
On-Demand	Pay per hour/second, zero commitment needed
Reserved Instances	Pay upfront — save up to 72% on costs
Spot Instances 🛠️	Use unused capacity — save up to 90%! Best deal in cloud

Free Tier

12 months FREE for new users to try AWS

🔗 AWS is better because it came **FIRST**, has the **MOST** services, **LARGEST** global network, **BIGGEST** community, and is trusted by the world's most critical organizations.

VANISH



CHAPTER 5 — What is IAM in AWS?



Simple Explanation — The Office Building

Imagine you are the OWNER of a big office building 🏢

Your building has these rooms:

💰 Finance Room 💻 Server Room
📁 Files Room 🔬 Research Room

Different employees get different ID cards:

- Receptionist → Can only enter the Lobby
- Accountant → Can only enter Finance Room
- IT Guy → Can only enter Server Room
- Manager → Can enter ALL rooms
- Guest → Cannot enter ANY room without permission



That ID card system = IAM (Identity and Access Management)



Technical Definition



Definition:

IAM (Identity and Access Management) is an AWS service that helps you control WHO can access WHAT in your AWS account.
It manages Authentication (who are you?) and Authorization (what can you do?).



4 Main Components of IAM

1. 👤 IAM USERS

A real person (like you or your colleague) who has their own username and password to log into AWS.

Example:

john@company.com → Developer (can only access EC2)
sara@company.com → DBA (can only access RDS Database)
admin@company.com → Admin (can access EVERYTHING)

2. 👥 IAM GROUPS

Instead of giving permissions to each person one by one — you create a GROUP and assign permissions to the whole group at once.

Developer Group → John, Priya, Ravi (all get same EC2 access)
DBA Group → Sara, Mohit (all get same RDS access)

Real Life: Like a department in a company — everyone in HR gets same HR file access.

3. 📄 IAM POLICIES

A JSON document (rulebook) that defines exactly what actions are Allowed or Denied on which AWS services.

4. 🎭 IAM ROLES

A temporary identity given to an AWS SERVICE or application — NOT a person. It is like a costume that a service wears to get temporary access.

Example: EC2 Server needs to read from S3 bucket.
→ Give EC2 an IAM Role with S3 read permission.
→ EC2 wears that role and accesses S3 temporarily.

Real Life: Like a contractor who gets a temporary access card for one day only — not permanent.

🏫 Real Life Analogy — The School Example

IAM Concept	School Analogy
IAM User	Each student/teacher has their own unique school ID card
IAM Group	Class 10A — all students in the group get same library access

IAM Policy	School rulebook — students can enter library, NOT staffroom
IAM Role	Substitute teacher gets temporary access for one day only
Root Account	The Principal — has access to absolutely EVERYTHING

IAM Authentication Methods

Method	Used For
Username + Password	Humans logging into the AWS Console (website)
Access Key + Secret Key	Programs or apps connecting to AWS via code (API)
MFA (Multi Factor Auth)	Extra security — Password + OTP from phone = double protection
IAM Role	AWS services talking to other AWS services — no password needed

Root Account vs IAM User

Feature	 Root Account	 IAM User
What is it?	The master account that created AWS	A sub-account created inside AWS
Access Level	100% access to EVERYTHING	Only what you give it
Used for	Initial setup only	Day-to-day work
Use daily?	☒ NO — Very dangerous!	☒ YES — Always use IAM
Real Life	Owner of the building	Employee with ID card

 **GOLDEN RULE:**
 NEVER use your Root Account for daily work.
 Always create IAM Users and use them instead!

✓ Why IAM is Important

Benefit	Explanation
 Security	Nobody gets more access than they need — ever
 Control	You decide exactly who does what in your AWS account
 Audit Trail	Every action is logged — know who did what and when
 Cost Protection	Limit access to expensive services to prevent misuse
 Free Service	IAM is completely FREE in AWS — zero extra charges

 **IAM = The security guard + ID card system of your AWS account — it controls WHO can enter and WHAT they can do inside.**



CHAPTER 6 — Practical Lab: S3 + IAM Hands-On

- ☒ Created an S3 Bucket
- ☒ Created an IAM User with auto-generated password
- ☒ Gave that IAM User access to the S3 Bucket
- ☒ Logged in as the IAM User
- ☒ Successfully viewed and accessed the S3 Bucket

PART 1 — Creating an S3 Bucket

Step	Action	Details
Step 1	Login to AWS	Go to console.aws.amazon.com and login with your Root/IAM account
Step 2	Go to S3	Search 'S3' in the top search bar and click on S3 service
Step 3	Create Bucket	Click the orange 'Create Bucket' button on the top right
Step 4	Enter Bucket Name	Give a unique name (e.g. my-first-bucket-yourname). Names must be globally unique!
Step 5	Choose Region	Select your nearest region (e.g. ap-south-1 for Mumbai, India)
Step 6	Block Public Access	Keep 'Block all public access' ON (default) — for security
Step 7	Create	Scroll down and click 'Create Bucket' — Done! Bucket is ready

What is S3?

S3 = Simple Storage Service

It is AWS's cloud storage — like Google Drive but for developers and companies.

You store files (called objects) inside containers (called buckets).

Bucket = Folder in the cloud

Object = File inside that folder

PART 2 — Creating an IAM User

Step	Action	Details
Step 1	Go to IAM	Search 'IAM' in the search bar and open the IAM service
Step 2	Click Users	In left sidebar click 'Users', then click 'Create User'
Step 3	Enter Username	Type a username (e.g. s3-read-user or john-dev)
Step 4	Enable Console Access	Check 'Provide user access to AWS Management Console'
Step 5	Auto-generate Password	Select 'Autogenerated password' — AWS creates a strong password
Step 6	Force Reset (optional)	Check 'User must create new password at next sign-in' for security
Step 7	Set Permissions	Click 'Attach policies directly' — search and select AmazonS3FullAccess
Step 8	Review & Create	Review all details and click 'Create User'
Step 9	Download Credentials	IMPORTANT: Download the .csv file with username + password NOW. You cannot see the password again!

IMPORTANT — Save Your Credentials!

After creating the IAM User, AWS shows the password ONLY ONCE. Download the CSV file immediately or copy the password and save it. If you lose it, you will have to reset the password manually.

PART 3 — Give IAM User Access to S3 Bucket

You already did this in Step 7 above by attaching AmazonS3FullAccess policy.

But here is what actually happened behind the scenes:

AWS created a POLICY (JSON rulebook) that says:

```
{ "Effect": "Allow", "Action": "s3:*", "Resource": "*" }
```

This means: This IAM User is ALLOWED to do ALL S3 actions on ALL buckets.

If you wanted to restrict to ONLY ONE bucket, you would change Resource to:
"Resource": "arn:aws:s3:::my-first-bucket-yourname/*"

PART 4 — Login as IAM User & View S3 Bucket

Step	Action	Details
Step 1	Get Login URL	Go to IAM > Dashboard. Copy the 'Account ID' or 'IAM Sign-In URL'
Step 2	Open IAM Login	The URL looks like: https://123456789.signin.aws.amazon.com/console
Step 3	Enter Credentials	Enter the IAM username and the autogenerated password you saved
Step 4	Reset Password	If forced reset was on, create a new password when prompted
Step 5	Go to S3	Search S3 in the console — you should be able to see the bucket!
Step 6	Verify Access	Click on the bucket and try to view contents — if you can see it, IAM is working perfectly!

What Actually Happened — Full Flow

Here is the complete flow of what you did today:

1. You (Root User) logged into AWS
- ↓
2. Created an S3 Bucket (storage container in cloud)
- ↓
3. Went to IAM and created a new User (john-dev)
- ↓
4. Attached AmazonS3FullAccess Policy to that user
- ↓
5. AWS gave john-dev a username + autogenerated password
- ↓
6. Logged in as john-dev using IAM login URL
- ↓
7. Opened S3 — Bucket was visible and accessible!
- ↓

8. IAM successfully CONTROLLED ACCESS to S3! ☒

What You Learned from This Practical

Concept	What the Practical Proved
IAM controls access	Without IAM policy, the user could NOT see the bucket
Least privilege works	You gave only S3 access — user cannot touch EC2 or RDS
Autogenerated password	AWS creates a strong secure password automatically
IAM login is separate	IAM users use a different URL than Root account
S3 is easy to use	Creating a bucket takes less than 2 minutes on AWS
Policies are powerful	One policy (AmazonS3FullAccess) gave complete S3 control

 **You completed your first real AWS practical — Created S3 Bucket + IAM User + Assigned Policy + Verified Access. This is exactly how companies manage cloud security in real world!**